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Classification of root systems

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0.1 Sylvester

Theorem 0.1.1 (Sylvester). *sylvester_criterion* $thm : posDef \text{ leading}_m inors_{pos}, thm : leading_m inors_{pos}, mp$
 $\times n \quad M$

1. M

2. $0 < k \leq n \quad M \text{ } k \quad (\text{ } k \times k \text{ })$

$M \in M_n(\mathbb{R}) \quad I = \{1, 2, \dots, n\} \quad (\text{Lean } 0, 1, \dots, n-1 \quad)$

0.1.1 Sylvester 1

Definition 0.1.2. *Matrix.leadingSubmatrix* $M_k \quad M \text{ } k$

$$\begin{aligned} X \quad |X| = m \text{ } e : X \rightarrow I \quad M_{e,e} \in M_m(\mathbb{R}) \\ (M_{e,e})_{ij} := M_{e(i), \text{ } e(j)} \end{aligned} \tag{1}$$

Lemma 0.1.3. *IsHermitian.submatrix_injective* $MM_{e,e}$

Lemma 0.1.4. *PosDef.submatrix_injective* $lem : IsHermitian.submatrix_injective e MM_{e,e}$

Proof. $??$
 $0 \neq x := {}^t(x_1, \dots, x_m) \in \mathbb{R}^m$

$$y_i = \begin{cases} x_j, & e(j) = i, \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

$y \in \mathbb{R}^n \text{ } y := {}^t(y_1, \dots, y_n)$

$$x^* M_{e,e} x = y^* M y. \tag{3}$$

$y = 0 \quad x = 0 \quad y \neq 0 \quad M \quad (??) \quad M_{e,e} \quad \square$

Sylvester $?? \implies ??$

Theorem 0.1.5. *posDef_leading_m_inors_pos_lem* : *PosDef.submatrix_injective* Mk
 $\leq n \quad \det M_k > 0$

Proof. $M_k \quad ?? \text{ } M_k \quad \square$

0.1.2 Sylvester 2

subsection $M \in M_{n+1}(\mathbb{R})$

Definition 0.1.6. $\text{lastCol } b := (b_1, \dots, b_n) \in \mathbb{R}^n \quad b_i = M_{i, n+1} \quad b \ M$

Definition 0.1.7. $\text{lastEntry } c \in \mathbb{R} \quad c = M_{n+1, n+1} \quad c \ M$

$$M = \left(\begin{array}{c|c} A & b \\ \hline {}^t b & c \end{array} \right) \quad (4)$$

Definition 0.1.8. $\text{schurComplement } s \in \mathbb{R} \ M$ **Schur complement**

$$s = c - {}^t b A^{-1} b. \quad (5)$$

Lemma 0.1.9. $\text{quadraticFormDecomp} M v := (x_1, \dots, x_n, t) \in \mathbb{R}^{n+1}$

$${}^t v M v = {}^t x A x + 2t({}^t b x) + c t^2 \quad (6)$$

Lemma 0.1.10. $\text{quadraticFormCompleteSquare} \text{lem} : \text{quadraticFormDecomp} M \Rightarrow {}^t v M v = {}^t (x + t A^{-1} b) A (x + t A^{-1} b) + s t^2 \quad (7)$

Lemma 0.1.11. $\text{schurComplementPos} M \text{det } M > 0 \quad M \text{ Schur complement } s$

$$0 < s = c - {}^t b A^{-1} b. \quad (8)$$

Proof. $\text{det } M = (\text{det } A) \cdot s \quad A \quad \text{det } A > 0 \quad \text{det } M > 0 \quad s > 0 \quad \square$

Lemma 0.1.12. $\text{posDef} \text{dotProduct} n \text{onneg} A \in M_k(\mathbb{R}) \quad x \in \mathbb{R}^k \quad {}^t x A x \geq 0$

Theorem 0.1.13. $\text{posDef} \text{ofLeadingSubmatrix} \text{posDef} \text{and} \text{detPos} \text{lem} : \text{schurComplementPos} \text{lem} : \text{posDef} \text{dotProduct} n \text{onneg} A \in M_k(\mathbb{R}) \quad x \in \mathbb{R}^k \quad {}^t x A x \geq 0$

Proof. $0 \neq v := (x_1, \dots, x_n, t) \in \mathbb{R}^{n+1} \quad {}^t v M v > 0$
 $t = 0 \quad A$

$${}^t v M v = {}^t x A x > 0 \quad (9)$$

$t \neq 0 \quad ??$

$${}^t v M v = {}^t (x + t A^{-1} b) A (x + t A^{-1} b) + s t^2 \quad (10)$$

1 A 2 $?? \ s > 0 \quad {}^t v M v > 0 \quad \square$

Lemma 0.1.14. $\text{leadingSubmatrix} \text{leadingSubmatrix} j \leq k \leq n \quad M \ k \quad j \quad M \ j$

$$(M_k)_j = M_j \quad (11)$$

Theorem 0.1.15. $\text{leadingMinorPos} \text{impliesPosDef} \text{thm} : \text{posDef} \text{ofLeadingSubmatrix} \text{posDef} \text{and} \text{detPos} \text{lem} : A \in M_n(\mathbb{R}) \quad k \leq n \quad 0 < k \quad 0 < M_k \quad M$

0.2

0.2.1 Cartan

Cartan $A_n, B_n, \dots, G_2$ Mathlib

Definition 0.2.1. CartanMatrix.A Cartan $A_n \in M_n(\mathbb{Z})$

$$A_n = \begin{pmatrix} 2 & -1 & 0 & 0 & \cdots & 0 \\ -1 & 2 & -1 & 0 & \cdots & 0 \\ 0 & -1 & 2 & -1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 2 \end{pmatrix} \quad (12)$$

Coxeter-Dynkin ??
R

Definition 0.2.2. CartanMatrix.ind_matrixY $\in M_n(R)$, $a, b \in R$ $M_{\text{ind}}(Y, a, b) \in M_{n+1}(R)$

$$M_{\text{ind}}(Y, a, b) = \left(\begin{array}{c|c} Y & \begin{matrix} 0 \\ \vdots \\ 0 \\ b \end{matrix} \\ \hline 0 & \cdots & 0 & a & 2 \end{array} \right). \quad (13)$$

Lemma 0.2.3. CartanMatrix.ind_{det}X $\in M_{n+2}(R)$, $Y \in M_{n+1}(R)$, $Z \in M_n(R)$, $a, b \in R$ $X = M_{\text{ind}}(Y, a, b)$, $Y_{n-1} = Z$

$$X = \left(\begin{array}{c|c} Y & \begin{matrix} 0 \\ \vdots \\ 0 \\ b \end{matrix} \\ \hline 0 & \cdots & 0 & a & 2 \end{array} \right) = \left(\begin{array}{c|c|c} Z & * & \begin{matrix} 0 \\ \vdots \\ 0 \end{matrix} \\ \hline * & * & b \\ \hline 0 & \cdots & 0 & a & 2 \end{array} \right) \quad (14)$$

0.3

graphics/positive_{definite}graph.pdf

Figure 1:

graphics/non-positive $_{definite_g}$ raph.pdf

Figure 2: